

REINFORCE++: Stabilizing Critic-Free Policy Optimization with Global Normalization

Anonymous authors
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Abstract

Reinforcement Learning from Human Feedback (RLHF) plays a crucial role in aligning Large Language Models (LLMs). The dominant algorithm, Proximal Policy Optimization (PPO), employs a critic network to estimate advantages, which introduces significant computational and memory overhead. To address this, a family of critic-free algorithms (e.g., GRPO, RLOO) has emerged. However, these methods typically rely on prompt-level (local) advantage normalization, which suffers from inaccurate advantage estimation, a tendency to overfit, and, as we show, is a theoretically biased estimator. To solve these challenges, we introduce REINFORCE++, a critic-free framework centered on Global Advantage Normalization. By normalizing advantages across the entire global batch rather than small, prompt-specific groups, our method yields a more stable estimator and, under standard assumptions, enjoys asymptotically vanishing normalization-induced bias as batch size increases. We introduce two variants: REINFORCE++, a computationally efficient algorithm ($k \geq 1$) for general-domain RLHF, and REINFORCE++_{w/ Baseline}, a group-sampling variant ($k > 1$) for complex reasoning tasks. Across our evaluated settings, each variant improves training stability and matches or exceeds strong critic-free baselines; in complex agentic settings, REINFORCE++_{w/ Baseline} is competitive with PPO.

1 Introduction

Reinforcement Learning from Human Feedback (RLHF) is a key technique for aligning Large Language Models (LLMs) with human values and preferences (Vemprala et al., 2023; Achiam et al., 2023; Ouyang et al., 2022; Shen & Zhang, 2024; Shen et al., 2025; Hu et al., 2024). Despite the emergence of non-RL alternatives like DPO (Rafailov et al., 2023), state-of-the-art applications such as ChatGPT/GPT-4 (Vemprala et al., 2023; OpenAI, 2023), Claude (Anthropic, 2023), and Gemini (Team et al., 2023) continue to rely on RL algorithms.

The dominant RL algorithm in this space is Proximal Policy Optimization (PPO) (Schulman et al., 2017). PPO employs an "Actor-Critic" architecture, where a dedicated critic network is trained to estimate the advantage function (i.e., the extent to which an action yields a higher expected return than the mean). However, this critic network introduces substantial computational overhead and memory requirements, making training prohibitively expensive and limiting large-scale model alignment on small-scale clusters (Shao et al., 2024).

To address PPO’s efficiency issues, a family of REINFORCE-based methods has emerged without the critic model, including ReMax (Li et al., 2023), RLOO (Ahmadian et al., 2024), and GRPO (Shao et al., 2024). These algorithms replace the critic network with an estimator that computes the advantage from multiple responses to the same prompt.

However, this critic-free approach introduces a new, critical challenge: advantage estimation. Methods like GRPO and RLOO use prompt-level (local) normalization, calculating the baseline and standard deviation only from the small group of responses generated for a single prompt. This approach suffers from several critical flaws. First, as demonstrated in Appendix Section A, the prompt-level advantage estimator is mathematically biased, as the numerator (centered reward) and the denominator (local standard deviation) are not

independent. Second, the advantage estimate is sensitive to the small sample size in the local group (e.g., $k = 4$ or $k = 8$), which leads to instability. If all samples receive similar rewards, the local $\text{std}(\cdot)$ approaches zero, causing the advantage to explode and destabilizing training. Finally, the method encourages overfitting to specific prompts, as the policy is optimized to "win" within its local group rather than achieving a globally high reward, which harms generalization.

We introduce two distinct variants of this framework, each tailored to a specific use case. REINFORCE++ is an efficient algorithm with $k \geq 1$. In its $k = 1$ configuration (detailed in Algorithm 1), it maximizes prompt diversity for general-domain RLHF. It can also be applied with $k > 1$ (Section 4.2), using the same global normalization principle. REINFORCE++_{w/ Baseline} is a specialized variant for complex tasks ($k > 1$) that benefits from group sampling. It fixes the flaws of GRPO by first subtracting the group mean (for reward reshaping) and then normalizing by the global standard deviation (for stability). It also employs a more stable k_2 for the KL estimator, as detailed in Appendix Section B. In summary, our contributions are as follows:

- We provide a theoretical proof that the prompt-level (local) advantage normalization used in methods like GRPO is a biased estimator (Appendix Section A).
- We propose REINFORCE++, a critic-free framework centered on Global Advantage Normalization, as a stable, efficient, and theoretically sound alternative.
- We present two specialized algorithms: We can use REINFORCE++ when $k \geq 1$ for efficient general RLHF and reasoning, and use REINFORCE++_{w/ Baseline} when $k > 1$ for robust complex agentic tasks.
- The experiments empirically demonstrate that our method REINFORCE++ achieves superior token efficiency and generalization when in out-of-distribution tasks. At the same time, REINFORCE++_{w/ Baseline} prevents overfitting and outperforms both GRPO and PPO in complex agentic reasoning tasks.

2 Background and Related Work

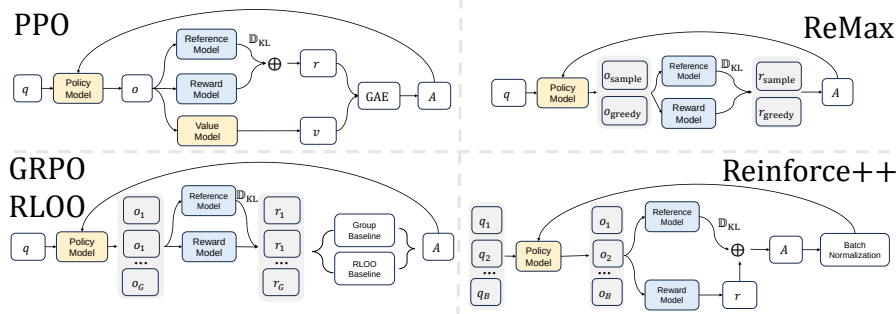


Figure 1: The comparison of PPO, ReMax, GRPO, RLOO, and REINFORCE++. REINFORCE++, as shown in the figure, removes the critic model and uses global batch normalization. REINFORCE++_{w/ Baseline} uses group sampling, similar to GRPO/RLOO, but replaces the local group baseline with a group-mean subtraction followed by global batch normalization.

2.1 PPO and Critic-Free RLHF

PPO optimizes LLMs by maximizing the following surrogate objective:

$$\mathcal{L}_{\text{PPO}}(\theta) = \mathbb{E}_{q \sim P(Q), o \sim \pi_{\theta_{\text{old}}}(O|q)} \left[\frac{1}{|o|} \sum_{t=1}^{|o|} \min \left(s_t(\theta) A_t, \text{clip}(s_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t \right) \right] \quad (1)$$

where $s_t(\theta) = \frac{\pi_{\theta}(o_t|q, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t|q, o_{<t})}$ is the probability ratio. The advantage A_t is typically calculated using Generalized Advantage Estimation (GAE) (Schulman et al., 2015):

$$A_{q, o_t} = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l} \quad (2)$$

where $\delta_{q, o_t} = r_t + \gamma V(o_{t+1}) - V(o_t)$ is the temporal difference error, and $V(\cdot)$ is the critic network.

Critic-free methods in Figure 1 remove the critic $V(\cdot)$ and compute the advantage A_t directly from rewards.

- ReMax (Li et al., 2023) uses a greedy decoding response $\hat{o}$ as the baseline: $A_{q, o_t} = r(o) - r(\hat{o})$.
- RLOO (Ahmadian et al., 2024) samples k responses and uses the mean of others as the baseline: $A_{q, o_t^{(i)}} = r(o^{(i)}) - \frac{1}{k-1} \sum_{j \neq i} r(o^{(j)})$.
- GRPO (Shao et al., 2024) also samples k responses but normalizes the advantage using the mean and standard deviation of the local group:

$$A_{q, o_t^{(i)}} = \frac{r(o^{(i)}) - \text{mean}(\{r(o^{(j)})\}_{j=1}^k)}{\text{std}(\{r(o^{(j)})\}_{j=1}^k) + \epsilon} \quad (3)$$

2.2 The Problem with Local Normalization

The core issue with methods like GRPO lies in Eq. (3). The usage of prompt-level (local) normalization is problematic for three reasons:

1. **Theoretical Bias:** As formally proven in Appendix Section A, the estimator is biased. Therefore, the numerator (centered reward) and the denominator (local standard deviation $\text{std}(\cdot)$) are not independent. The denominator’s value is correlated with the rewards in the small group, introducing a systematic error in the advantage estimate (Mai et al., 2025).
2. **Practical Instability:** In practice, the group size k is small (e.g., 4 or 8). If all sampled responses for a prompt happen to get similar rewards, the local $\text{std}(\cdot)$ approaches zero, causing the advantage to explode. It results in high variance and unstable training.
3. **Task Overfitting:** The policy is rewarded for being "better than other samples from the same prompt," not for being "globally good." It can lead to overfitting on simple prompts where it’s easy to generate diverse rewards, while failing to improve on complex prompts.

3 Method

Our method addresses the instability of critic-free RLHF by replacing biased local normalization with stable global normalization. We propose two variants for different use cases.

3.1 Global Normalization

The primary REINFORCE++ algorithm is designed for general-purpose RLHF. In its $k = 1$ configuration, it is designed for maximum efficiency and prompt diversity.

It optimizes the PPO objective in Eq. (1) but redefines the advantage A_{q,o_t} . We use the standard PPO reward formulation, which incorporates the kl-style KL penalty directly into the reward:

$$A_{q,o_t} = r(o_{1:T}, q) - \beta \cdot \sum_{i=t}^T \text{KL}(i) \quad (4)$$

where $\text{KL}(t) = \log \left(\frac{\pi_{\theta_{\text{old}}}(o_t|q, o_{<t})}{\pi_{\text{ref}}(o_t|q, o_{<t})} \right)$.

The key innovation is our normalization strategy. Instead of a non-existent (for $k = 1$) or biased local norm, we use Global Advantage Normalization (Andrychowicz et al., 2020):

$$A_{q,o_t}^{\text{norm}} = \frac{A_{q,o_t} - \text{mean}(A|A \in \mathcal{D}_{\text{batch}})}{\text{std}(A|A \in \mathcal{D}_{\text{batch}}) + \epsilon} \quad (5)$$

The global normalization is the core of REINFORCE++. As the global batch size $\mathcal{D}_{\text{batch}}$ is typically large (e.g., 1024 or more), the $\text{mean}(\cdot)$ and $\text{std}(\cdot)$ converge to stable constants. It makes the gradient estimator effectively less biased (as $N \rightarrow \infty$) and robust to outliers, drastically improving training stability (see Appendix Section 3.5). The algorithm for the efficient $k = 1$ case is detailed in Algorithm 1.

3.2 Local Baseline

For more complex tasks, such as multi-step reasoning, sampling multiple responses per prompt ($k > 1$) can be beneficial (Yue et al., 2025). To this end, we introduce REINFORCE++_{w/ Baseline}. This variant combines the benefits of group-sampling with the stability of global normalization. The advantage calculation is a two-step process:

- **Group Mean Subtraction for Reshaping:** We first subtract the group mean reward. It serves as a local baseline, reshaping rewards to be robust to different reward scales (e.g., 0/1 vs. -1/1).

$$A'_{q,o_t} = R_{q,o_t} - \text{mean}_{\text{group}}(R_{q,o_t}) \quad (6)$$

- **Global Batch Normalization for Stability:** We then normalize this initial advantage using the global batch statistics, not the unstable local group statistics.

$$A_{q,o_t}^{\text{norm}} = \frac{A'_{q,o_t} - \text{mean}_{\text{batch}}(A')}{\text{std}_{\text{batch}}(A') + \epsilon} \quad (7)$$

This combination avoids the bias and instability inherent in the local standard deviation in GRPO.

Furthermore, this variant employs a separate KL loss term for regularization. We adopt the k_2 estimator. As shown in Appendix Section B, the k_2 estimator provides a stable, unbiased gradient for the Reverse KL divergence, unlike the k_3 estimator (used in GRPO), which is an unstable approximation (Liu et al., 2025a). The final objective is:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}}(A^{\text{norm}}) - \lambda \cdot \mathcal{J}_{k_2 \text{ as loss}}(\theta) \quad (8)$$

where $\mathcal{J}_{k_2 \text{ as loss}}(\theta) = \mathbb{E}[\frac{1}{2}(\log \frac{\pi_{\theta}}{\pi_{\text{ref}}})^2]$.

3.3 Relationship with PPO

REINFORCE++_{w/ Baseline} can be viewed as a simplified and more stable variant of PPO. It is formally equivalent to a PPO agent where: (1) The critic network is removed; (2) The GAE parameters are set to $\lambda = 1$ and $\gamma = 1$; and (3) A two-step global batch normalization is used as the baseline instead of a learned value function.

3.4 Implementation Tricks

Token-level Advantage The advantage A_{q,o_t}^{norm} is computed at the token level. For $t < T$ (where T is the sequence length), the advantage is set to 0. For the final token $t = T$, the advantage is the normalized reward A_{q,o_T}^{norm} , which is the standard practice in RLHF for LLMs.

Batch Construction For REINFORCE++ ($k = 1$), a global batch of size N comprises N distinct prompts. For REINFORCE++_{w/ Baseline} ($k = 4$), a global batch of size $N = 1024$ might contain $1024/4 = 256$ unique prompts. The global mean/std are computed over all N samples.

Mini-Batch Updates To enhance training efficiency, we implement mini-batch updates with the following characteristics:

- **Batch Processing:** Data is processed in smaller, manageable chunks rather than full-batch updates.
- **Multiple Updates:** Each mini-batch allows for multiple parameter updates, improving convergence rates.
- **Stochastic Optimization:** Introduces beneficial randomness for better generalization.

Reward Normalization and Clipping We implement comprehensive reward processing to stabilize training:

- **Normalization:** Standardizes rewards using z-score normalization (our global normalization) to mitigate outliers.
- **Clipping:** Constrains reward values within predefined bounds to avoid instability.
- **Scaling:** Applies appropriate scaling factors for numerical stability during updates.

3.5 Why Use Global Batch Normalization?

We observe that as $N \rightarrow \infty$, the denominator of the estimator converges to the constant σ , and the bias in the numerator vanishes. The insight underlies our decision to adopt global batch normalization in both REINFORCE++ and REINFORCE++_{w/ Baseline}, as the global batch size (N_{global}) is typically much larger (e.g., 1024) than the group batch size (N_{group} , e.g., 4 or 8) in practice. As $N_{global} \rightarrow \infty$, the sample statistics μ_b and σ_b converge to true constants, making the estimator effectively unbiased.

3.6 Summary

To address these challenges, we propose REINFORCE++, a method centered on a simple yet powerful idea: Global Advantage Normalization, without a critic. Instead of normalizing within a prompt’s local group, REINFORCE++ normalizes the advantage function across the entire global training batch. The overall approach is:

- **Theoretical Stability:** As the global batch size grows (e.g., $N = 1024$), the batch mean and standard deviation converge to constants, resulting in an effectively unbiased (bias vanishes as $N \rightarrow \infty$) and low-variance advantage estimator (see Appendix Section 3.5).
- **Training Efficiency:** It retains the critic-free architecture, significantly reducing computational and memory overhead compared to PPO.
- **Strong Generalization:** Using a global baseline prevents overfitting to specific prompts and encourages a more robust policy.

Algorithm 1 REINFORCE++ when $k = 1$

Require: Initial policy model π_{ref} , reward models R , task prompts $\mathcal{D}$

- 1: policy model $\pi_\theta \leftarrow \pi_{ref}$
- 2: for step = $1, \dots, M$ do
- 3: Sample a batch $\mathcal{D}_{batch}$ from $\mathcal{D}$
- 4: Update the old policy model $\pi_{old} \leftarrow \pi_\theta$
- 5: Sample output ($k = 1$) $o \sim \pi_{old}(\cdot | q)$ for each question $q \in \mathcal{D}_{batch}$
- 6: Compute rewards r_i for each sampled o_i using R
- 7: Compute advantage A_{q,o_t} via Eq. (4)
- 8: Normalize advantages globally across $\mathcal{D}_{batch}$ to get A_{q,o_t}^{norm} via Eq. (5)
- 9: for iteration = $1, \dots, k$ do
- 10: Update π_θ by maximizing the objective in Eq. (1) using A_{q,o_t}^{norm}
- 11: end for
- 12: end for

Ensure: π_θ

4 Experiments

We evaluate our two algorithms, REINFORCE++ and REINFORCE++_{w/ Baseline}, in their respective target domains using the OpenRLHF framework (Hu et al., 2024).

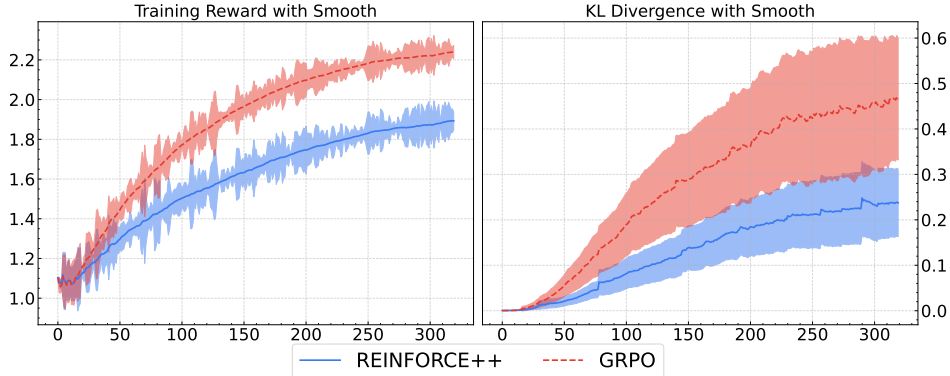


Figure 2: Comparison of smoothed Training Reward and KL Divergence. REINFORCE++ ($k = 1$) achieves strong rewards with significantly more stable (lower) KL divergence, avoiding GRPO’s reward-hacking behavior.

4.1 General RLHF

First, we evaluate the REINFORCE++ in its $k = 1$ configuration, focusing on general-domain RLHF, where efficiency and prompt diversity are key.

Experimental Setup We use an instruction-following policy model (Llama-3-8B-SFT) and refine it using a Bradley-Terry reward model (Bai et al., 2022) trained on $\sim 700K$ human preference pairs. The policy is trained on 20,000 diverse prompts. We compare REINFORCE++ ($k = 1$) with other critic-free methods: GRPO ($k = 4$), RLOO ($k = 4$), and ReMax ($k = 1 + 1$).

Experimental Results As shown in Table 1, the single-sample REINFORCE++ ($k = 1$) achieves a score of 46.7, statistically tied with the group-sampling GRPO (46.8). However, it produces shorter responses (832 tokens vs. 860), yielding a higher, more efficient per-token score (0.0561). This result indicates that for general tasks, group sampling ($k > 1$) is unnecessary and may be suboptimal.

	Score	Length	Per Token
REINFORCE++ (k=1)	46.7	832	0.0561
GRPO (k=4)	46.8	860	0.0544
RLOO (k=4)	44.6	866	0.0515
ReMax (k=1+1)	45.1	805	0.0560

Table 1: Comparison on Chat-Arena-Hard (Li et al., 2024). The single-sample REINFORCE++ ($k = 1$) achieves a top-tier score while being more token-efficient than the group-sampling GRPO.

Results Analysis Figure 2 shows the training dynamics. GRPO’s reward rises quickly, but its KL divergence also increases rapidly, suggesting it is ”hacking” the reward model. In contrast, REINFORCE++ shows a more stable reward increase with a much smaller KL divergence. It highlights the stability of global normalization and its higher ”KL-to-reward” conversion efficiency, avoiding the reward hacking seen in local normalization methods, which is often linked to length exploitation (Dubois et al., 2024).

4.2 Reasoning Experiments

Next, we evaluate REINFORCE++ (group-sample, $k > 1$) in its target domain: complex reasoning, where group sampling is standard, and reward signals are often sparse (e.g., 0/1). We compare it directly with GRPO ($k > 1$).

4.2.1 Long Reasoning Task

We use rule-based rewards (RLVR) in mathematical reasoning settings (Guo et al., 2025; Seed et al., 2025).

Analysis on Small-Scale Datasets To test for overfitting, we trained on only 30 questions from AIME-24 and evaluated on AIME-25. As shown in Table 2, GRPO (local norm) achieves a near-perfect 95.0% on the training set but completely fails on the test set (0.0 Pass@1), demonstrating catastrophic overfitting. In contrast, REINFORCE++ with global norm, while scoring lower on the training set (71.0), shows significantly better generalization (2.5 Pass@1, 40.0 Pass@16).

Pass@N	AIME-24 (Train)	AIME-25 (Test)	
	N = 1	N = 1	N = 16
GRPO	95.0	0.0	0.4
REINFORCE++	71.0	2.5	40.0

Table 2: Comparison on a small training dataset. GRPO (local norm) overfits completely, while REINFORCE++ (global norm) generalizes.

Figure 3 confirms this. GRPO (left) immediately overfits, mastering the training questions in just a few steps. REINFORCE++ (right) learns more gradually, enabled by the stable global normalization signal.

Logical Reasoning (K&K Puzzles) We also tested on the Knights and Knaves (K&K) puzzles (Xie et al., 2025; 2024), where difficulty increases with the number of ”people.” Figure 4 shows that while GRPO is competitive on easy tasks (2-3 people), its performance collapses on harder, OOD tasks (8 people). REINFORCE++ is more robust, outperforming GRPO on all tasks with four or more people and achieving a much higher average score (62.1 vs. 55.7).

RL from Zero Setting Finally, we trained a Qwen2.5-Math-Base model from zero on MATH dataset splits (Guo et al., 2025). Table 3 shows that REINFORCE++ again achieves bet-

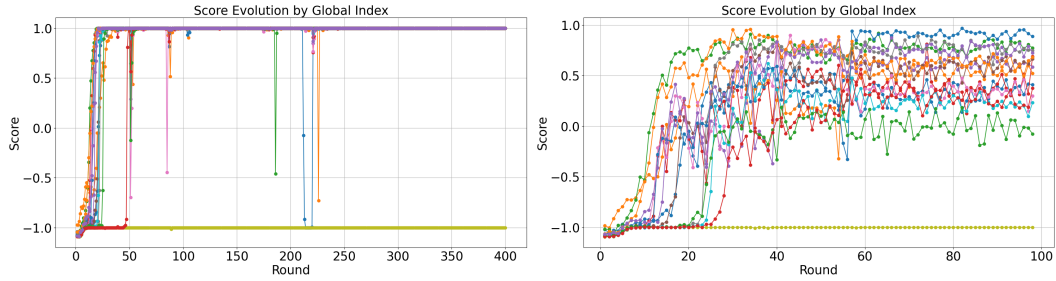


Figure 3: Training curves on a small prompt dataset. Left: GRPO (local norm) immediately overfits. Right: REINFORCE++ (global norm) learns more stably.

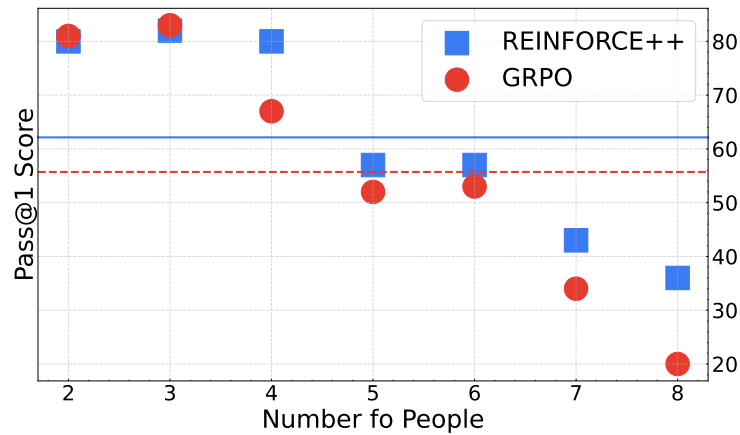


Figure 4: Comparison on logic benchmarks. REINFORCE++'s global normalization provides better robustness to increasing task difficulty and OOD challenges (8 people).

ter OOD generalization on the more challenging AIME-24 and AMC-23 datasets, while remaining competitive on the in-distribution MATH-500 test set.

Pass@N	AIME-24 (OOD) N = 8	AMC-23 (OOD) N = 8	MATH-500 (ID) N = 1
GRPO	18.96	59.22	73.00
REINFORCE++	21.04	60.47	72.00

Table 3: Comparison on RL from Zero. REINFORCE++ shows superior OOD performance.

4.3 Multi-step Reinforcement Learning

Finally, to test performance in a complex, multi-step environment, we use the zero-shot agent setup from Mai et al. (2025), in which a Qwen 2.5 Base 7B model must learn to use Python tools to solve mathematical problems. It is a challenging scenario where group sampling ($k > 1$) and reward reshaping are crucial (see Appendix Section 5).

Experimental Setup We adhere to the training and evaluation protocols of the ZeroTIR environment Mai et al. (2025). The backbone model is Qwen 2.5 Base 7B, trained using OpenRLHF Hu et al. (2025) with datasets from ORZ Hu et al. (2024) and DAPO Yu et al. (2025). We assess performance on benchmarks including AIME 2024 Jia (2024), AIME-2025 math ai (2025), HMMT FEB-2024/2025, and CMIMC, using the average@32 metric.

Experimental Result As shown in Table 4, REINFORCE++_{w/ Baseline} achieves the highest average accuracy (24.10) across all benchmarks. It significantly outperforms both GRPO (22.58) and the full-critic PPO (21.85). It demonstrates that our stable, critic-free approach is highly effective for complex agentic tasks, surpassing even the heavyweight PPO algorithm. The combination of group-mean reshaping, stable global normalization, and the correct ‘k2’ KL loss proves superior.

Algorithm	AIME 24	AIME 25	HMMT 2025	HMMT 2024	CMIMC	Avg
GRPO (local norm)	31.66	21.87	16.97	17.70	24.68	22.58
PPO (critic-based)	30.20	21.66	15.00	18.43	23.95	21.85
RF++-Baseline (global norm)	30.83	27.18	17.91	18.95	25.62	24.10

Table 4: Performance Comparison on complex tool-use benchmarks (average@32). REINFORCE++_{w/ Baseline} outperforms both GRPO (local norm) and PPO (critic-based) models.

5 Best Practices

5.1 General Principle

This paper introduces two algorithms: REINFORCE++ and REINFORCE++_{w/ Baseline} [n], with a key focus on selecting the appropriate algorithm based on task conditions.

Empirical evidence from the open-source community suggests that REINFORCE++_{w/ Baseline} [n] is particularly effective for sample filtering or more complex scenarios. For instance, in the multi-turn tool-calling setting, a high proportion of void (i.e., non-informative or incorrect) samples can destabilize training. In such cases, incorporating a baseline by subtracting the intra-group mean reward significantly improves training stability by effectively filtering out void samples. Moreover, this automatic reward reshaping reduces the need for designing complex reward structures. For example, REINFORCE++_{w/ Baseline} [n] supports both 0/1 and -1/1 reward schemes, while REINFORCE++ performs best with symmetric rewards, such as -1/1 in RLVR tasks.

In contrast, for general-domain tasks where prompt diversity and efficiency are paramount, or for tasks where obtaining multiple reward signals for distinct responses is challenging—such as training with Process-Supervised Reward Models (PRMs) or online real-time sampling—we recommend using REINFORCE++ ($k = 1$). Our experiments (Section 4.1) show that it provides superior OOD generalization in these settings.

5.2 Third-Party Validation

The core principle of REINFORCE++ , global advantage normalization, has been independently validated and adopted in several subsequent large-scale reasoning systems, confirming its stability and effectiveness.

LitePPO (Liu et al., 2025c), which effectively combines REINFORCE++_{w/ Baseline} with a token-level loss, conducted experiments demonstrating that the global standard deviation is superior to the local standard deviation used in GRPO. Their results indicate more stable training and better generalization, which aligns with our findings.

ScaleRL (Khatri et al., 2025) performed a detailed ablation study on advantage estimation methods in large-scale (16,000 GPU-hour) experiments. They directly compared batch-level normalization in REINFORCE++ with prompt-level normalization in GRPO. Their findings concluded that batch-level normalization was "slightly superior in both compute efficiency and final performance," confirming the advantages of global normalization at scale.

DLER (Liu et al., 2025b) found that when using truncation to control the output length of an LLM, batch-wise (global) normalization remains stable while group-wise (local) normalization shows declining accuracy. The study provides further evidence that global normalization is more robust to variations in training conditions and reward landscapes.

6 Conclusion

In this paper, we present REINFORCE++ , a critic-free RLHF framework designed to enhance training stability by addressing fundamental flaws in existing methods. We identified that prior critic-free algorithms, such as GRPO, rely on prompt-level (local) advantage normalization, which we prove to be theoretically biased (Appendix Section A) and demonstrate to be practically unstable.

Our solution, Global Advantage Normalization, normalizes advantages across the entire batch, providing a stable and effectively unbiased estimator (whose bias vanishes as batch size increases). We proposed two variants tailored for different needs:

1. **REINFORCE++**: For general-domain RLHF, we show the single-sample ($k = 1$) approach is highly efficient and achieves superior OOD generalization. We also show that its application with $k > 1$ is robust for reasoning tasks (Section 4.2).
2. **REINFORCE++_{w/ Baseline}**: For complex reasoning and agent tasks, this variant combines group-mean reshaping with stable global normalization and a theoretically sound k_2 estimator for KL.

Our empirical results, supported by independent validation from multiple third-party systems, confirmed the effectiveness of this specialized approach. REINFORCE++ demonstrated state-of-the-art generalization in general RLHF. REINFORCE++_{w/ Baseline} showed dramatic improvements in stability, preventing overfitting in low-data regimes and outperforming both GRPO and full-critic PPO in complex, long-horizon tool-use tasks. This work demonstrates that a theoretically sound, stable approach without a critic model can be more efficient and effective than traditional PPO.

AI Use Statement

Generative AI tools were used to assist with the LaTeX template migration, formatting checks, and compilation diagnostics. They were not used to generate experimental data, fabricate results, or make scientific claims. The authors reviewed all AI-assisted work and take responsibility for the final content of this paper, including its text, claims, and artifacts.

Ethics Statement

Reinforcement learning is increasingly used to steer large language models by optimizing them against a reward signal, including verifiable, programmatic rewards (RLVR) for domains such as math, code, and tool use. The main potential positive impact of this line of work is improved scalability and reproducibility of post-training: verifiable rewards can reduce reliance on human labeling and make optimization targets easier to audit and replicate. By improving the stability and efficiency of critic-free optimization under sparse and brittle rewards, our method can help make RLVR-style training more reliable.

Potential risks include reward gaming when verifiers are incomplete or mis-specified, and dual-use concerns if improved tool-use or problem-solving capabilities are deployed without safeguards. We recommend evaluating beyond verifier scores (e.g., robustness and red-teaming), stress-testing verifiers for exploitability, and using conservative deployment controls (e.g., access restrictions and monitoring) for systems trained with RL.

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Appendix

A Proof: The GRPO Advantage Estimator is Biased

A.1 Assumptions and Settings

We observe N rewards r_i for a prompt, and assume the true baseline is θ , such that

$$r_i = \theta + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, N,$$

with all advantage value ϵ_i independent. Define

$$\bar{\epsilon} = \frac{1}{N} \sum_{j=1}^N \epsilon_j, \quad D = \sqrt{\frac{1}{N} \sum_{j=1}^N (\epsilon_j - \bar{\epsilon})^2}, \quad A_i = \frac{\epsilon_i - \bar{\epsilon}}{D}.$$

We will prove the following:

Theorem A.1. For any finite $N \geq 2$, the advantage estimator A_i is biased:

$$\mathbb{E}[A_i \mid \epsilon_i] \neq \epsilon_i.$$

Proof. We can prove our findings with the following steps:

Step 1: Numerator Bias We rewrite the numerator:

$$\epsilon_i - \bar{\epsilon} = \left(1 - \frac{1}{N}\right) \epsilon_i - \frac{1}{N} \sum_{j \neq i} \epsilon_j.$$

Since the ϵ_j for $j \neq i$ are zero-mean and independent of ϵ_i ,

$$\mathbb{E}[\epsilon_i - \bar{\epsilon} \mid \epsilon_i] = \left(1 - \frac{1}{N}\right) \epsilon_i.$$

Step 2: Denominator Depends on ϵ_i

(a) Compute $\mathbb{E}[D^2 \mid \epsilon_i]$: By definition,

$$D^2 = \frac{1}{N} \sum_{j=1}^N (\epsilon_j - \bar{\epsilon})^2 = \frac{1}{N} \sum_{j=1}^N \epsilon_j^2 - \bar{\epsilon}^2.$$

Since

$$\bar{\epsilon} = \frac{1}{N} \left(\epsilon_i + \sum_{j \neq i} \epsilon_j \right),$$

and conditioning on ϵ_i keeps the ϵ_j , $j \neq i$, i.i.d. $\mathcal{N}(0, \sigma^2)$, we obtain:

$$\mathbb{E} \left[\sum_{j=1}^N \epsilon_j^2 \mid \epsilon_i \right] = \epsilon_i^2 + (N-1)\sigma^2,$$

$$\begin{aligned} \mathbb{E}[\bar{\epsilon}^2 \mid \epsilon_i] &= \frac{1}{N^2} \mathbb{E} \left[\left(\epsilon_i + \sum_{j \neq i} \epsilon_j \right)^2 \mid \epsilon_i \right] \\ &= \frac{1}{N^2} \left(\epsilon_i^2 + 2\epsilon_i \cdot \underbrace{\mathbb{E} \left[\sum_{j \neq i} \epsilon_j \right]}_0 + \mathbb{E} \left[\left(\sum_{j \neq i} \epsilon_j \right)^2 \right] \right) \\ &= \frac{\epsilon_i^2 + (N-1)\sigma^2}{N^2}. \end{aligned}$$

Subtracting:

$$\begin{aligned}\mathbb{E}[D^2 \mid \epsilon_i] &= \frac{1}{N}(\epsilon_i^2 + (N-1)\sigma^2) - \frac{\epsilon_i^2 + (N-1)\sigma^2}{N^2} \\ &= \underbrace{\frac{(N-1)^2}{N^2}\sigma^2}_{\alpha} + \underbrace{\frac{N-1}{N^2}\epsilon_i^2}_{\beta} = \alpha + \beta\epsilon_i^2.\end{aligned}\quad (9)$$

(b) $g(\epsilon_i)$ is Not Constant: Let $g(\epsilon_i) = \mathbb{E}[1/D \mid \epsilon_i]$ and $\mu(\epsilon_i) = \mathbb{E}[D^2 \mid \epsilon_i] = \alpha + \beta\epsilon_i^2$.

Using Taylor expansion of $f(x) = 1/\sqrt{x}$ around $x_0 = \mu(\epsilon_i)$:

$$f(x) = \frac{1}{\sqrt{x_0}} - \frac{1}{2} \frac{x - x_0}{x_0^{3/2}} + \frac{3}{8} \frac{(x - x_0)^2}{x_0^{5/2}} + O((x - x_0)^3)$$

Taking the conditional expectation:

$$\begin{aligned}g(\epsilon_i) &= \mathbb{E}[1/D \mid \epsilon_i] = \mathbb{E}\left[\frac{1}{\sqrt{D^2}} \mid \epsilon_i\right] = \mathbb{E}[f(D^2) \mid \epsilon_i], \quad \text{where } f(x) = \frac{1}{\sqrt{x}} \\ &\approx f(\mu(\epsilon_i)) + f'(\mu(\epsilon_i)) \cdot \mathbb{E}[D^2 - \mu(\epsilon_i) \mid \epsilon_i] + \frac{f''(\mu(\epsilon_i))}{2} \cdot \mathbb{E}[(D^2 - \mu(\epsilon_i))^2 \mid \epsilon_i] \\ &= \frac{1}{\sqrt{\mu(\epsilon_i)}} - \frac{1}{2\mu(\epsilon_i)^{3/2}} \cdot \underbrace{\mathbb{E}[D^2 - \mu(\epsilon_i) \mid \epsilon_i]}_{=0} + \frac{3}{8\mu(\epsilon_i)^{5/2}} \cdot \text{Var}(D^2 \mid \epsilon_i) \\ &= \frac{1}{\sqrt{\mu(\epsilon_i)}} + \frac{3}{8} \cdot \frac{\text{Var}(D^2 \mid \epsilon_i)}{\mu(\epsilon_i)^{5/2}}\end{aligned}$$

Since $\mu(\epsilon_i) = \alpha + \beta\epsilon_i^2$ with $\beta > 0$, the first term alone shows that $g(\epsilon_i)$ depends on ϵ_i^2 and hence is not constant.

Step 3: Putting It Together Decomposing A_i ,

$$A_i = \frac{\epsilon_i - \bar{\epsilon}}{D} = \left(1 - \frac{1}{N}\right) \frac{\epsilon_i}{D} - \left(\frac{1}{N} \sum_{j \neq i} \epsilon_j\right) \cdot \frac{1}{D}.$$

For fixed ϵ_i , the conditional distribution of $\sum_{j \neq i} \epsilon_j$ is symmetric about zero, while $1/D$ is always positive. Thus:

$$\mathbb{E}\left[\left(\frac{-1}{N} \sum_{j \neq i} \epsilon_j\right) \cdot \frac{1}{D} \mid \epsilon_i\right] = 0.$$

It follows that

$$\mathbb{E}[A_i \mid \epsilon_i] = \left(1 - \frac{1}{N}\right) \epsilon_i \cdot \mathbb{E}\left[\frac{1}{D} \mid \epsilon_i\right] = \left(1 - \frac{1}{N}\right) \epsilon_i \cdot g(\epsilon_i).$$

Step 4: Concluding the Bias If A_i were unbiased, we would have:

$$\left(1 - \frac{1}{N}\right) g(\epsilon_i) \equiv 1 \quad \Rightarrow \quad g(\epsilon_i) \equiv \frac{N}{N-1},$$

which contradicts Step 2. Therefore, for any finite $N \geq 2$,

$$\mathbb{E}[A_i \mid \epsilon_i] \neq \epsilon_i.$$

Hence A_i is a biased estimator.

□

B KL Penalty Design

In non-critic algorithms like REINFORCE++_{w/ Baseline}, a KL divergence term is often added as a separate loss to constrain the policy π_θ from deviating too far from a reference policy π_{ref} . In practice, this expectation is estimated using samples, and three familiar estimators are used.

Given the importance ratio $\delta(y) = \pi_{ref}(y|\cdot)/\pi_\theta(y|\cdot)$ (where π_θ is the sampling policy and π_{ref} is the reference):

- $k_1(y) = -\log \delta(y) = \log \frac{\pi_\theta(y|\cdot)}{\pi_{ref}(y|\cdot)}$
- $k_2(y) = \frac{1}{2}(\log \delta(y))^2 = \frac{1}{2} \left(\log \frac{\pi_{ref}(y|\cdot)}{\pi_\theta(y|\cdot)} \right)^2$
- $k_3(y) = \delta(y) - 1 - \log \delta(y)$

We analyze which of these (k_1 , k_2 , or k_3) is the correct choice when used as a separate loss term.

Theoretical Reverse KL Gradient We are training with samples from π_θ , so we are approximating the **Reverse KL (RKL)**, $D_{KL}(\pi_\theta || \pi_{ref})$. The practical policy gradient for RKL is (more details are in (Liu et al., 2025a)):

$$\nabla_\theta \mathcal{J}_{RKL}(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot|x)} \left[\left(\log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)} \right) \nabla_\theta \log \pi_\theta(y|x) \right]$$

k_1 Analysis We can **exclude k_1** as a loss term. Its gradient does not depend on π_{ref} and thus provides no constraining effect. The loss $\mathcal{J}_{k_1} = \mathbb{E}_{y \sim \pi_\theta} [\log \pi_\theta - \log \pi_{ref}]$ differentiates to $\nabla_\theta \mathcal{J}_{k_1} = \mathbb{E}_{y \sim \pi_\theta} [\nabla_\theta \log \pi_\theta]$, as the π_{ref} term vanishes. (Note: k_1 **is** used inside the reward for REINFORCE++, but not as a separate loss term).

k_2 Analysis The k_2 estimator provides a gradient that is **equivalent** to the theoretical Reverse KL gradient.

The k_2 loss is (noting that $(\log \frac{\pi_\theta}{\pi_{ref}})^2 = (\log \frac{\pi_{ref}}{\pi_\theta})^2$):

$$\mathcal{J}_{k_2 \text{ as loss}}(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot|x)} \left[\frac{1}{2} \left(\log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)} \right)^2 \right]$$

Differentiating the term inside the expectation gives:

$$\begin{aligned} \nabla_\theta \left[\frac{1}{2} \left(\log \frac{\pi_\theta}{\pi_{ref}} \right)^2 \right] &= \left(\log \frac{\pi_\theta}{\pi_{ref}} \right) \cdot \nabla_\theta \left(\log \frac{\pi_\theta}{\pi_{ref}} \right) \\ &= \left(\log \frac{\pi_\theta}{\pi_{ref}} \right) \cdot \nabla_\theta (\log \pi_\theta - \log \pi_{ref}) \\ &= \left(\log \frac{\pi_\theta}{\pi_{ref}} \right) \nabla_\theta \log \pi_\theta \end{aligned}$$

When we take the expectation over $y \sim \pi_\theta$, the gradient of the k_2 loss becomes:

$$\mathbb{E}_{y \sim \pi_\theta} \left[\left(\log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)} \right) \nabla_\theta \log \pi_\theta(y|x) \right]$$

This perfectly matches the practical RKL gradient. Therefore, k_2 is the theoretically correct and stable choice.

k_3 Analysis The k_3 gradient is problematic. It is theoretically an estimator for the **Forward KL** ($KL(\pi_{ref} || \pi_\theta)$), not the Reverse KL. There is a mismatch: we are sampling from π_θ , but the Forward KL gradient requires sampling from π_{ref} .

This method, used in GRPO, suffers from two major issues:

- Extremely High Variance: When $\pi_\theta(y)$ becomes tiny for a sample y where $\pi_{ref}(y)$ is moderate, the importance weight $\frac{\pi_{ref}(y)}{\pi_\theta(y)}$ in the gradient estimator explodes, leading to "infinite variance".
- Numerical Instability: It requires computing p/q via $\exp(\log p - \log q)$, which is prone to overflow.

The result explains why methods using k require frequent resetting of π_{ref} to prevent the policies from diverging and the k_3 estimator from becoming unstable.

Conclusion Based on this analysis, the k_2 estimator is the optimal choice for the KL loss term, as it accurately and stably estimates the Reverse KL gradient. We therefore use the k_2 estimator in our REINFORCE++_{w/ Baseline} algorithm.